

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO3: ~~Analyze~~ relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)

Activity Tool : MCQ Test (Low)

Assessment Tool Weightage: 30%

QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	5

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: $A \twoheadrightarrow B$ and $B \twoheadrightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

I M.Vishnu Vardhan Reddy192411053 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Vishnu Vardhan Reddy/

1. Which of the following is true about 1NF?

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic
- (d) Eliminates transitive dependencies

Answer: (c) All attributes must be atomic

2. A relation is in 2NF if it is in 1NF and:

- (a) Has no transitive dependencies
- (b) Has no partial dependencies on primary key
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Answer: (b) Has no partial dependencies on primary key

3. Which normal form deals with multi-valued dependencies?

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF

Answer: (d) 4NF

4. BCNF is a stricter version of:

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) 4NF

Answer: (c) 3NF

5. Functional dependency $X \rightarrow Y$ means:

- (a) Y determines X
- (b) X determines Y
- (c) X and Y are unrelated
- (d) Both determine each other

Answer: (b) X determines Y

6. Which of the following anomaly is NOT associated with unnormalized relations?

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly

Answer: (d) Retrieval anomaly

7. A superkey is:

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

Answer: (b) Any set of attributes that uniquely identifies tuples

8. If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A
- (d) Independently determined

Answer: (c) Transitively dependent on A

9. 5NF is also known as:

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form
- (d) Element Normal Form

Answer: (c) Project-Join Normal Form

10. Lossless-join decomposition ensures:

- (a) No data is lost during decomposition
- (b) All FDs are preserved
- (c) The relation is in BCNF
- (d) No null values exist

Answer: (a) No data is lost during decomposition

Answer Key

1. (c) All attributes must be atomic
2. (b) Has no partial dependencies on primary key
3. (d) 4NF
4. (c) 3NF
5. (b) X determines Y
6. (d) Retrieval anomaly
7. (b) Any set of attributes that uniquely identifies tuples
8. (c) Transitively dependent on A
9. (c) Project-Join Normal Form
10. (a) No data is lost during decomposition